Zero dimension for the real exceptional set and vanishing 3-adic nesting constants

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Abstract

Let $\Sigma \subset \mathbb{Z}_3$ be the set of 3-adic integers with digits in $\{0,1\}$, and let F_k consist of the points $x \in \mathbb{Z}_3$ for which $2^n x \in \Sigma$ at least k times. We prove $\lim_{k\to\infty} \dim_{\mathrm{H}} F_k = \lim_{k\to\infty} \dim_{\mathrm{P}} F_k = 0$. For each $\delta > 0$, some finite k admits a countable compact cover of F_k whose members have upper box dimension at most δ. The arithmetic input is a uniform finite extraction theorem: sufficiently many distinct powers of two yield templates of any fixed arity, with common rational 3-adic unit coefficients and disjoint positive distance sets. Its proof uses a uniform unit-equation bound, finite Ramsey theory, valuation-minimal minors, and a two-dimensional residue-field obstruction.

We also prove that the positive real numbers λ for which $2^n\lambda$ has a full ternary expansion omitting 2 for infinitely many n form a set of Hausdorff and packing dimension zero, again with countable compact covers of arbitrarily small upper box dimension. This is the untruncated real exceptional set of Lagarias's Conjecture 1.4. Its proof retains a separate infinite-set extraction and a finite-word transfer: real truncations satisfy affine 3-adic constraints with shifts in fixed finite sets. Both arguments use the finite causal-container theorem of version 1.0.0, whose complete frozen proof is reproduced unchanged. The required two-translate sumset capacity estimate is also proved here. Neither conclusion excludes the point 1 or proves countability or emptiness of an exceptional set.

Relation to version 1.0.0. The previous version [4] proved zero Hausdorff and packing dimension for the 3-adic infinite-visit set E_2. The present version strengthens that conclusion to uniform finite-visit compact covers and vanishing nesting constants, and adds the real exceptional-set theorem. The earlier E_2 conclusion is recovered as a corollary. The two new arguments are given in full and share the unchanged finite causal-container appendix. No implication from a bare 3-adic dimension formula to the real theorem is assumed.

1 The 3-adic finite-visit problem

We use the metric $|x-y|_3$ on $\mathbb{Z}_3$, with $v_3(3) = 1$ and $v_3(0) = +\infty$. Put

$$\Sigma = \left\{ \sum_{j\ge 0} x_j 3^j : x_j \in \{0,1\} \right\}, \qquad F_k = \{x \in \mathbb{Z}_3 : \#\{n \ge 0 : 2^n x \in \Sigma\} \ge k\}.$$

The visits in this definition occur at distinct nonnegative integer times. Set

$$\Gamma_{\mathrm{H}} = \lim_{k\to\infty} \dim_{\mathrm{H}} F_k, \qquad \Gamma_{\mathrm{P}} = \lim_{k\to\infty} \dim_{\mathrm{P}} F_k.$$

These limits exist because the sets decrease. The Hausdorff limit is the dyadic nesting constant considered in the study of the exceptional set by Lagarias and by Abram, Bolshakov and Lagarias [1, 2]. We retain the arbitrary-visit convention throughout: in particular $F_1 = \bigcup_{n\ge 0} 2^{-n}\Sigma$, rather than literally Σ.

Theorem 1.1. *For every $\delta > 0$ there is an integer $J \geq 1$ and a countable family of compact subsets $K_m \subset \mathbb{Z}_3$ such that*

$$F_J \subseteq \bigcup_{m \geq 1} K_m, \qquad \overline{\dim}_{\mathrm{B}} K_m \leq \delta \quad (m \geq 1).$$

Consequently

$$\boxed{\Gamma_{\mathrm{H}} = \Gamma_{\mathrm{P}} = 0.}$$

The distinction from an infinite-visit argument is the order of quantifiers. The cardinality J will be fixed before any visiting times are given. The exponent spans of the extracted templates need not be bounded uniformly. They are fixed separately on each member of the countable cover before its digit length tends to infinity.

The analytic input is the finite causal-container theorem in Appendix A of the frozen release v1.0.0 of [4]. The real theorem is stated in Theorem 7.1; its proof begins in Section 7 and uses a separate arithmetic extraction. We state exactly the portion used in Section 5 and reproduce its complete source unchanged in Appendix A. The new arithmetic extraction is proved in full below. In the 3-adic finite-visit argument, neither the extraction from an infinite set in [4] nor a uniform bound on the spans of finite templates is used.

2 Uniform finite extraction: statement and incidence bound

For a finite set of nonnegative integers P write

$$\Delta(P) = \{v - u : u, v \in P,\ u < v\}.$$

These are sets, so equal distances within one P are counted only once.

Theorem 2.1 (Uniform finite templates)**.** *Fix integers $r \geq 3$, $K \geq 1$, and $q \geq 1$. There is a finite integer $J = J(r, K, q)$ with the following property. Every set of J distinct powers 2^n, $n \geq 0$, contains K distinct members γ_i and admits rational units*

$$\alpha, A_1, \ldots, A_{r-2}, u_1, \ldots, u_K \in \mathbb{Q} \cap \mathbb{Z}_3^{\times}$$

such that

$$c_i := \gamma_i/\alpha = 1 + \sum_{t=1}^{r-2} A_t 3^{e_{ti}} + u_i 3^{E_i} \qquad (1 \leq i \leq K). \tag{1}$$

The positions

$$P_i = \{0, e_{1i}, \ldots, e_{r-2,i}, E_i\}$$

are in increasing order, every successive gap is at least q, and $\Delta(P_i) \cap \Delta(P_j) = \varnothing$ for $i \neq j$. The bound J is independent of the original powers and of all exponent spans. The coefficient α is a virtual unit anchor: it need not be one of the original powers.

We prove this theorem by induction on the number of common monomial coordinates. At a stage with $d \geq 1$ coordinates the data are

$$\gamma_i = 2^{n_i} \quad \text{distinct}, \qquad 0 = e_{0i} < e_{1i} < \cdots < e_{d-1,i}, \tag{2}$$

$$\gamma_i = \sum_{t=0}^{d-1} a_t 3^{e_{ti}} + \varepsilon_i, \quad a_t \in \mathbb{Q} \cap \mathbb{Z}_3^{\times}, \quad v_3(\varepsilon_i) \geq e_{d-1,i} + q. \tag{3}$$

Successive gaps in (2) are at least q, and every column $e_{ti} - e_{si}$, $s < t$, is injective as i varies. For $d = 1$ the last condition is empty.

We use the uniform theorem of Evertse, Schlickewei and Schmidt [3, Theorem 1.1]. In characteristic zero, an equation $a_1z_1+\cdots+a_hz_h=1$, with nonzero coefficients and variables in a subgroup of $(L^\times)^h$ of rank ρ, has at most

$$\exp((6h)^{3h}(\rho+1))$$

nondegenerate solutions. Here nondegenerate means that no nonempty subsum on the left vanishes. The theorem applies over an algebraic closure of $\mathbb{Q}_3$ and hence to the solutions considered here.

Lemma 2.2 (Uniform incidence bound)**.** *Suppose the γ_i are distinct powers of two and every difference column $e_{ti}-e_{si}$ is injective. For any fixed coefficients $c_0,\dots,c_{d-1}\in\mathbb{Q}_3$, the equation*

$$\gamma_i=\sum_{t=0}^{d-1}c_t3^{e_{ti}} \tag{4}$$

holds for at most

$$B_d=\binom{d}{2}+\left\lceil\exp((6d)^{3d}(d+2))\right\rceil$$

indices i.

Proof. Delete zero coefficients. If no coefficient remains, there is no solution. For each pair of remaining terms, equality of valuations fixes one value of $e_{ti}-e_{si}$ and occurs on at most one row. Excluding at most $\binom{d}{2}$ rows, all terms on the right of (4) have different valuations; no nonempty subsum of them can vanish.

Choose one remaining index t_0 and divide by $c_{t_0}3^{e_{t_0i}}$:

$$c_{t_0}^{-1}\gamma_i3^{-e_{t_0i}}-\sum_{t\neq t_0}\frac{c_t}{c_{t_0}}3^{e_{ti}-e_{t_0i}}=1. \tag{5}$$

A vanishing subsum not containing the γ_i term is impossible by the distinct valuations. A vanishing subsum containing that term would, on using (4), force a nonempty complementary subsum of the original right-hand side to vanish; the complement includes the t_0 term. Thus (5) is nondegenerate.

Its variable tuple belongs to the image of $\langle2\rangle\times\langle3\rangle^d$, a group of rank at most $d+1$. Two rows with the same tuple have $2^{n_i-n_j}=3^{e_{t_0i}-e_{t_0j}}$, hence $n_i=n_j$ by multiplicative independence. Thus rows give distinct solutions. The uniform unit-equation bound gives the stated number. If only one coefficient remains, multiplicative independence gives at most one row directly. □

3 The extension lemma

Lemma 3.1 (One additional coordinate)**.** *For every $d\geq1$ and $L\geq1$ there is a finite $N_d(L)$ such that, from $N_d(L)$ rows satisfying (2)–(3) and the injectivity conditions above, one can retain L rows and find rational units $b_0,\dots,b_{d-1}$ with*

$$\gamma_i=\sum_{t=0}^{d-1}b_t3^{e_{ti}}+u_i3^{E_i},\quad u_i\in\mathbb{Q}\cap\mathbb{Z}_3^\times,\quad E_i\geq e_{d-1,i}+q, \tag{6}$$

such that every column E_i-e_{ti}, $0\leq t<d$, is injective. The previous exponent columns do not change. The threshold $N_d(L)$ can be chosen independently of q, the coefficients and the exponents.

3.1 Growth order and valuation-minimal minors

Give the rows any linear order. Color a pair $i < j$ by the signs of all differences

$$(e_{tj} - e_{sj}) - (e_{ti} - e_{si}) \qquad (s < t).$$

None is zero. Finite Ramsey theory gives an arbitrarily large homogeneous subset, provided the original set is sufficiently large. The numbers $\Delta_t(i,j) = e_{tj} - e_{ti}$ have the same strict order for every pair $i < j$: their pairwise comparisons are fixed, and they are transitive since they compare actual numbers. Temporarily relabel the coordinates $1, \dots, d$ in this growth order, so that

$$\Delta_1(i,j) < \cdots < \Delta_d(i,j) \qquad (i < j). \tag{7}$$

This temporary order need not agree with the increasing order of exponents within a row.

For $s < t$ and $i < j$, (7) gives

$$e_{si} + e_{tj} > e_{sj} + e_{ti}. \tag{8}$$

It follows that every square submatrix of $(3^{e_{ti}})$ has exactly one determinant term of least valuation. Indeed, exchanging an assignment which is not order-reversing strictly lowers its valuation by (8). The unique minimum pairs increasing row indices with decreasing growth-coordinate indices. In particular every d-by-d coordinate minor is nonzero, and its valuation equals that minimum permutation cost.

Adjoin γ_i to the coordinate column and write

$$v_i = (3^{e_{1i}}, \dots, 3^{e_{di}}, \gamma_i)^{\mathsf{T}} \in \mathbb{Q}_3^{d+1}.$$

Color each $(d+1)$-subset according to whether its augmented determinant is zero. An all-zero subset of size greater than B_d is impossible: its augmented matrix would have rank at most d, while the coordinate matrix has rank d, forcing the γ_i row to be one fixed linear combination of the coordinate rows, contrary to Lemma 2.2. A second Ramsey refinement therefore gives as many rows as needed with every augmented $(d+1)$-minor nonzero.

3.2 Canonical determinant valuations

For an increasing $(d+1)$-tuple $I = (i_1, \dots, i_{d+1})$ put

$$D_I = \det(v_{i_1}, \dots, v_{i_{d+1}}), \qquad w_I = v_3(D_I).$$

Consider an increasing $(d+2)$-tuple. For each slot $h \in \{1, \dots, d+1\}$, compare its two $(d+1)$-subtuples obtained by keeping either of the adjacent indices in positions $h, h+1$ and keeping every other index. Color this comparison $t \in \{1, \dots, d\}$ if the change in w equals the change in e_t at the variable slot; otherwise color it G. At most one coordinate color is possible by (7). There are at most $(d+1)^{d+1}$ combined colors. Another finite Ramsey refinement makes the comparison color at each slot independent of the chosen indices.

If one slot has color G, fix the d other indices as pivots, with a large consecutive group of variable indices in that slot. For these variables, each column

$$w_{\text{pivots},i} - e_{ti} \qquad (1 \le t \le d) \tag{9}$$

is injective. This is the situation needed for interpolation below. We now prove that a G slot must exist on a sufficiently large homogeneous subset.

Suppose every slot has a coordinate color t_h. The graph of increasing $(d+1)$-tuples, with an edge for a change in only one coordinate, is connected. Along such an edge the value

$w_I - \sum_h e_{t_h,i_h}$ is unchanged, because the union of the two tuples is an increasing $(d+2)$-tuple of the type just colored. Consequently there is one constant C with

$$w_{i_1,\dots,i_{d+1}} = C + \sum_{h=1}^{d+1} e_{t_h,i_h} \tag{10}$$

for every increasing tuple in the homogeneous subset.

For a coordinate t and an increasing $(d+2)$-tuple I, the cofactor identity is

$$\sum_{l=1}^{d+2} (-1)^l 3^{e_{t,i_l}} D_{I\setminus i_l} = 0. \tag{11}$$

Its nonzero summands cannot have a unique least valuation. Color I by a selected pair (l_t, m_t), $l_t < m_t$, of minimum-valuation positions for each t. A further Ramsey refinement fixes these pairs for all tuples. This uses at most $\binom{d+2}{2}^d$ colors.

Write A_l for the valuation of the lth summand of (11). For a fixed t and its fixed pair $l < m$, the identity $A_m - A_l = 0$, after inserting (10), has the following slot contributions:

$$\begin{cases} e_{t_l,i_l} - e_{t,i_l}, & \text{at slot } l, \\ e_{t_j,i_j} - e_{t_{j-1},i_j}, & l < j < m, \\ e_{t,i_m} - e_{t_{m-1},i_m}, & \text{at slot } m. \end{cases} \tag{12}$$

All other contributions cancel. Hold the other slots fixed and give one slot two possible indices. This is possible for every slot in an ordered set with at least $d+3$ elements. A difference of two distinct coordinate functions is injective by hypothesis, so it cannot have the same value for these two choices. Hence all the labels in each slot function of (12) coincide, and

$$t_l = t_{l+1} = \cdots = t_{m-1} = t. \tag{13}$$

In particular each of the d coordinate labels appears in the word $t_1, \dots, t_{d+1}$. Exactly one label appears twice and all the others appear once.

This word is nonincreasing in the growth order. Indeed, adjacent summand valuations for coordinate t satisfy

$$A_{j+1} - A_j = \Delta_t(i_j, i_{j+1}) - \Delta_{t_j}(i_j, i_{j+1}). \tag{14}$$

Suppose $t_h = a < b = t_{h+1}$. If a appeared only once, its fixed minimum pair, by (13), would be $h, h+1$. But the next difference in (14) is negative, making A_{h+2} smaller than that minimum. Thus a must be the repeated label. If b appeared only once, its minimum pair would be $h+1, h+2$, while the preceding difference in (14) is positive, making A_h smaller than that minimum. Thus b must also repeat, a contradiction. Therefore the word has the form

$$d, d-1, \dots, t_*, t_*, \dots, 1. \tag{15}$$

3.3 A two-dimensional residue-field obstruction

Let the repeated label t_* occupy slots $h, h+1$. Fix $h-1$ pivot indices before a group of five variable indices, and $d-h$ pivots after it. There are $d-1$ pivots in total, so $d+4$ available indices suffice. Their augmented columns are independent, as follows by extending them to a nonzero d-by-d coordinate minor. Quotient $\mathbb{Q}_3^{d+1}$ by their span. This is a two-dimensional vector space; let u_i denote the images of the five variable columns.

In any fixed coordinates on the quotient, its pair determinants are the corresponding augmented determinants times one fixed nonzero factor (up to signs). By (10) and (15),

$$v_3 \det(u_i, u_j) = C' + e_{t_*,i} + e_{t_*,j}.$$

The vectors $z_i = 3^{-e_{t_*,i}} u_i$ therefore have all pair determinants nonzero and of the same valuation. There can be at most four such vectors. To see this without any normalization assumption on their norms, use z_1, z_2 as a basis and write $z_i = a_i z_1 + b_i z_2$ for $i \geq 3$. Comparison with the first two vectors gives $a_i, b_i \in \mathbb{Z}_3^\times$. Comparison between $i, j \geq 3$ gives $a_i b_j - a_j b_i \in \mathbb{Z}_3^\times$. Thus the residues of a_i/b_i are distinct nonzero elements of $\mathbb{F}_3$, of which there are only two. There are at most two vectors beyond the basis, contradicting the five variables. The assumption that all slots have coordinate colors is impossible. A G slot exists.

3.4 Interpolation and preservation of the gap

Restore the original within-row exponent order $0 = e_{0i} < \cdots < e_{d-1,i}$. Fix the d pivots of a G slot. Their coordinate matrix is invertible. Let b_t interpolate the γ_i on the pivots, and put $\delta_t = b_t - a_t$. The data for δ_t are the errors ε_i.

By Cramer's rule, $v_3(\delta_t) \geq q$ for every t. In fact, the denominator determinant has valuation exactly its minimum permutation cost, by (8). A term of a numerator determinant replaces one entry $3^{e_{ti}}$ in the corresponding denominator term by ε_i, whose valuation is at least $\max_s e_{si} + q \geq e_{ti} + q$. Every numerator term has valuation at least the denominator's minimum cost plus q. Cancellation can only increase this valuation. The b_t remain units; they are rational, since both the coordinate matrix and its interpolation data are rational.

For a variable index put

$$R_i = \gamma_i - \sum_{t=0}^{d-1} b_t 3^{e_{ti}}, \qquad E_i = v_3(R_i).$$

The augmented determinant formula for an interpolation residual gives $E_i = w_{\text{pivots},i} - C_0$ for one fixed constant C_0. All residuals are nonzero, and (9) shows that every $E_i - e_{ti}$ is injective.

It remains to discard rows on which $E_i < e_{d-1,i} + q$. On such a row the old ε_i cannot attain the least valuation in

$$R_i = \varepsilon_i - \sum_t \delta_t 3^{e_{ti}}.$$

If two *nonzero* error terms tie for the finite least valuation, that equality fixes one value of the corresponding difference column; there is at most one such row per unordered pair of coordinates. These rows number at most $\binom{d}{2}$. On every other bad row exactly one error term, say t, is least, giving

$$E_i - e_{ti} = v_3(\delta_t).$$

Injectivity of this residual column permits at most one such row for each t, hence at most d more. Deleting at most $d + \binom{d}{2}$ rows gives (6) and proves Lemma 3.1, subject only to choosing the finite cardinality thresholds, made explicit next.

3.5 Uniform cardinality bookkeeping

Let $\mathcal{R}_k(m;c)$ be any finite Ramsey number for c colors on k-subsets and monochromatic size m; use the usual one-color convention. For desired output L, put

$$\begin{aligned} b_d &= d + \binom{d}{2}, \qquad M = L + b_d, \\ H &= \max\left\{ d+M,\ \mathcal{R}_{d+2}\left(d+4; \binom{d+2}{2}^{d}\right)\right\}, \\ P &= \mathcal{R}_{d+2}\left(H; (d+1)^{d+1}\right), \\ Q_0 &= \mathcal{R}_{d+1}\left(\max\{P, B_d+1\}; 2\right), \\ N_d(L) &= \mathcal{R}_2\left(Q_0; 2^{\binom{d}{2}}\right). \end{aligned}$$

The refinements are respectively the growth comparisons, the nonzero augmented minors, and the fixed slot colors. A G slot leaves at least $H - d \geq L + b_d$ variable indices: take the required pivots at the two ends and all intermediate indices as variables. Discarding b_d bad rows leaves L. If there is no G slot, the minimum-pair refinement leaves $d+4$ indices, sufficient for every local slot variation and the five-variable quotient contradiction. This supplies a finite threshold depending only on d, L. In particular, no valuation depth or exponent span occurs in it.

4 Iteration and disjoint distance sets

Proof of Theorem 2.1. There are $U_q = 2\cdot 3^{q-1}$ unit residue classes modulo 3^q. Pigeonhole sufficiently many original powers into one such class and choose an ordinary unit representative a_0. Then $\gamma_i = a_0 + \varepsilon_i$, with $v_3(\varepsilon_i) \geq q$, provides the $d = 1$ starting stage.

Apply Lemma 3.1. If another common coordinate is needed, write its residual $R_i = u_i 3^{E_i}$ and retain one residue class of the units u_i modulo 3^q, represented by a unit a_d. Adjoin the common monomial $a_d 3^{e_{di}}$ with $e_{di} = E_i$. The remaining error has valuation at least $e_{di} + q$. All former difference columns are unchanged, and all new ones are injective by the extension lemma. Thus the induction hypotheses hold with $d+1$ common coordinates.

At $d = r-1$ perform one final extension, leaving its terminal unit coefficient free. There are then r positions, every successive gap is at least q, and each of the $c = \binom{r}{2}$ positive difference columns is injective. Form the graph on rows in which two rows are adjacent if their distance sets intersect. A fixed row has at most c distances. For each of these distances, each of the c difference columns attains it in at most one row. Thus the graph has maximum degree at most c^2. From $K(c^2+1)$ rows a greedy independent-set selection retains K pairwise disjoint distance sets. This argument allows repeated distances within an individual template.

For clarity, all input sizes may be fixed backwards. Start with $L_{r-1} = K(c^2+1)$ and require $N_{r-1}(L_{r-1})$ rows at that stage. For each preceding extension choose an output size U_q times the next stage's required input size, and apply the corresponding N_d. Finally multiply the $d = 1$ required input size by U_q for the initial residue pigeonhole. This is a finite integer $J(r, K, q)$ depending only on the displayed parameters.

The final common coefficient at exponent zero is a rational unit α. Divide the identity by α to obtain (1). All coefficients and terminal residuals are rational, because every interpolation used rational data. No bound on their heights or on the exponent spans has been asserted. □

5 A finite entropy estimate for weakly separated templates

For $z \in \mathbb{Z}_3$ let $[z]_j \in \{0, \dots, 3^j - 1\}$ denote its residue modulo 3^j, and define the length-q window

$$B_j^{(q)}(z) = \frac{[z]_{j+q} - [z]_j}{3^j}.$$

Let $\Sigma_L = \{\sum_{j=0}^{L-1} x_j 3^j : x_j \in \{0,1\}\}$. All entropies and logarithms below use the natural logarithm.

We first specify the external finite result. This is Appendix A, "Exact theorem", of [4].

Theorem 5.1 (Finite causal-container input)**.** *Let $r \geq 3$, $V_0 = \{0, \dots, N-1\}$, and Ω be an alphabet of size Q. A nonempty finite family $\mathcal{W} \subseteq \mathcal{X}^L$ determines letters $Z_j(x) \in \Omega$; these maps need not be local. Let a fixed family of ordered positional edges $e = (j_0 < \cdots < j_{r-1})$ have relations $\mathcal{R}_e(x_{<j_0}) \subseteq \Omega^r$ depending only on the indicated input prefix, and suppose every word satisfies every relation. Assume that distinct edges with the same source have disjoint future position sets and that each unordered pair of positions belongs to at most C edges.*

For positive integer thresholds $T_{\mathrm{s}}, T_2, \dots, T_{r-1}$ put $D = CQ^{r-3}$ and

$$\theta = Q\left[\frac{D\sum_{j=2}^{r-1} T_j + 1}{T_{\mathrm{s}}} + \frac{1}{T_2} + D\sum_{j=3}^{r-1} \frac{1 + \sum_{l=2}^{j-1} T_l}{T_j}\right]. \tag{16}$$

If $\theta \leq 1/2$, each word has a fingerprint $F(x)$ and nonempty assigned alphabets $A_j(x) \subseteq \Omega$ such that $Z_j(x) \in A_j(x)$, $A_j(x)$ is determined by $(x_{<j}, F(x))$, and, outside at most

$$Q\left(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j\right) N \tag{17}$$

edges, the full Cartesian product *of their assigned alphabets is contained in $\mathcal{R}_e(x_{<j_0})$. For every distribution on $\mathcal{W}$,*

$$H(F) \leq N\big[h_{\mathrm{bin}}(\theta) + \theta \log Q\big], \quad h_{\mathrm{bin}}(t) = -t\log t - (1-t)\log(1-t).$$

The proof of Theorem 5.1 is reproduced in Appendix A from the cited frozen source. The remaining argument, including its application to unrestricted ternary inputs and rational units, is given here.

Proposition 5.2 (Finite-word bound)**.** *Choose $b \geq 1$ so that $3^b > 2(r-1)$ and put $q = b+1$. Suppose* (1) *has successive position gaps at least this q and pairwise disjoint positive distance sets, as in Theorem 2.1. Set*

$$q = b+1, \quad Q = 3^{(r-1)q}, \quad C = \binom{r}{2}, \quad D = CQ^{r-3}, \quad R = \max_i E_i.$$

For $L \geq q$ let W_L be the set of all ternary input words modulo 3^L for which $[c_i X]_L \in \Sigma_L$ for every $1 \leq i \leq K$; no infinite extendability is required. Set $N = L - q + 1 = L - b$. With thresholds as in Theorem 5.1, define

$$\epsilon_F = h_{\mathrm{bin}}(\theta) + \theta \log Q, \qquad \eta = \frac{Q(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j)}{K}.$$

If $\theta \leq 1/2$, then

$$\log |W_L| \leq N\left[\epsilon_F + \frac{\log 2}{r} + \eta \log 3\right] + (R+b)\log 3. \tag{18}$$

5.1 Ordinary carries and the positional graph

For a source j and a template i with $v = j + E_i < N$, consider the r window positions

$$j, \quad v - e_{r-2,i}, \ldots, v - e_{1i}, \quad v.$$

At these positions use respectively the windows of X, the corresponding $A_t X$, and X again. The joint letter available at every position is

$$Z_j(X) = (B_j^{(q)}(X), B_j^{(q)}(A_1 X), \ldots, B_j^{(q)}(A_{r-2} X)). \tag{19}$$

It takes values in an alphabet of size Q.

At position v, the identity for $c_i X$ has r summands: X, the $r-2$ terms $3^{e_{ti}} A_t X$, and $3^{E_i} u_i X$. Their ordinary nonnegative residues below position v sum to a carry $h \in \{0, \ldots, r-1\}$ into that position. The terminal summand has window, modulo 3^q,

$$u_i B_j^{(q)}(X) + D_{ij}(X_{<j}), \qquad D_{ij} = \frac{3^{E_i} u_i [X]_j - [3^{E_i} u_i [X]_j]_v}{3^v}.$$

Here $D_{ij} \in \mathbb{Z}_3$ depends only on the input digits before j. Consequently the corresponding edge relation is

$$u_i B_j^{(q)}(X) + \sum_{t=1}^{r-2} B_{v-e_{ti}}^{(q)}(A_t X) + B_v^{(q)}(X) + D_{ij} + h \in \Sigma_q \pmod{3^q} \tag{20}$$

for some $0 \le h \le r-1$. This derivation is valid for rational 3-adic units, with all residues interpreted in $\mathbb{Z}_3$. It imposes no restriction $X \in \Sigma$.

Every future displacement from j is a positive distance in P_i. Thus different templates have disjoint future sets at the same source. If two positions belong to an edge, their distance belongs to exactly one of the template distance sets, and there are at most $\binom{r}{2}$ choices of roles in that template. The source is then fixed. The graph therefore satisfies the two hypotheses of Theorem 5.1 with the stated C. The relations (20) are causal, as their only extra parameter is D_{ij}.

5.2 The one-digit Cartesian-product bound

Freeze the low $b = q-1$ digits of each of the r windows. Let each remaining high digit range in a nonempty subset of $\mathbb{F}_3$. Suppose the full Cartesian product of these choices satisfies (20), allowing the target and carry to vary with the choice. For a target $y \in \Sigma_q$, its low part satisfies $0 \le [y]_b \le R_b = (3^b - 1)/2$. Hence

$$[y]_b - h \in [-(r-1), R_b].$$

This interval has length $R_b + r - 1 < 3^b$. For the fixed residue modulo 3^b determined by the frozen inputs and D_{ij}, it therefore contains at most one representative. After subtracting that representative and dividing by 3^b, the sum of the high-digit factors, with the first multiplied by the unit $u_i \bmod 3$, lies in a single translate of $\{0, 1\} \subset \mathbb{F}_3$.

No factor can have size three. Two factors of size at least two would already have sumset $\mathbb{F}_3$. Thus at most one factor has size two and all the others have size one. The product of the factor cardinalities is at most two, or equivalently their log-cardinalities sum to at most $\log 2$.

5.3 Conditional supports, incidence, and entropy

Apply Theorem 5.1 to the uniform distribution on W_L, using (19) and (20). (The set is nonempty because it contains zero.) For each realized word and $0 \le j < N$ put

$$C_j = \operatorname{supp}(X_{j+b} \mid X_{<j+b}, F), \qquad w_j = \log |C_j| \le \log 3.$$

Under this conditioning the low b digits of every window of X or $A_t X$ starting at j are fixed. Multiplication by a unit maps the high input digit bijectively onto the high output digit, so the projected support of the appropriate window has cardinality $|C_j|$. Moreover it lies in the relevant projection of the assigned alphabet A_j: every word in this finer conditional atom has the same $(X_{<j}, F)$ as the realized word.

On a good edge the full-product conclusion of Theorem 5.1 permits these projected supports to be combined, although the supports at different positions come from different prefix atoms. Freezing the low digits to their realized values and using the preceding Cartesian-product bound gives

$$\sum_{j \in e} w_j \le \log 2.$$

A bad edge instead costs at most $r \log 3$. Let E_{bad} be the number of bad edges for the realized word. There are at most KN edges. For a given template, each role covers all positions except at most $E_i \le R$ boundary positions. Summing incidences over all roles and templates gives

$$rK \sum_{j=0}^{N-1} w_j - rKR \log 3 \le \sum_e \sum_{j \in e} w_j \le KN \log 2 + rE_{\mathrm{bad}} \log 3.$$

The estimate remains valid when some templates have no edges because $E_i \ge N$, since the stated boundary loss then dominates the missing incidences. Dividing by rK and using $E_{\mathrm{bad}} \le \eta KN$ yields

$$\sum_j w_j \le N \frac{\log 2}{r} + \eta N \log 3 + R \log 3.$$

The windows' high input digits are precisely the positions $b, b+1, \dots, L-1$. The first b digits cost at most $b \log 3$. The chain rule and the support bound on conditional entropy give

$$H(X \mid F) \le b \log 3 + \mathbb{E} \sum_j w_j.$$

Finally $H(X) \le H(F) + H(X \mid F)$, and the fingerprint bound is $H(F) \le N\epsilon_F$. Since X is uniform on W_L, this proves (18) and Proposition 5.2.

6 Compact covers and the limit of finite-visit dimensions

Fix $r \ge 3$. Choose b, $q = b + 1$, Q, C, D as above. For an integer $u \ge 2$ set

$$T_j = u^{2^{j-1}-1} \ (2 \le j \le r-1), \qquad T_{\mathrm{s}} = u^{2^{r-1}-1}, \qquad K = u^{2^r}. \tag{21}$$

Every ratio occurring in (16) tends to zero as $u \to \infty$. Also $Q(T_{\mathrm{s}} + D \sum_j T_j)/K \to 0$. Thus

$$\theta \longrightarrow 0, \qquad \epsilon_F \longrightarrow 0, \qquad \eta \longrightarrow 0 \quad \text{for fixed } r. \tag{22}$$

For any one extracted finite template system $\mathcal{D}$ define

$$K_{\mathcal{D}} = \{X \in \mathbb{Z}_3 : c_i X \in \Sigma \text{ for } 1 \le i \le K\}.$$

It is compact. Its number of occupied length-L cylinders is at most $|W_L|$. Proposition 5.2, applied with the fixed span R of this system and then $L \to \infty$, gives

$$\overline{\dim}_{\mathrm{B}} K_{\mathcal{D}} \le s_{r,u} := \frac{\log 2}{r \log 3} + \frac{\epsilon_F}{\log 3} + \eta. \tag{23}$$

The fact that W_L contains all locally valid finite words only makes this upper bound larger; an assumption of infinite extendability is unnecessary.

Proof of Theorem 1.1. Given $\delta > 0$, first choose r so that $\log 2/(r \log 3) < \delta/2$. Then choose a finite u sufficiently large that $\theta \le 1/2$ and $\epsilon_F/\log 3 + \eta < \delta/2$. Fix the associated K, q and $J = J(r, K, q)$ from Theorem 2.1.

For any $x \in F_J$, choose J distinct true visit slopes $\gamma = 2^n$ and apply that finite theorem. The normalized variable $X = \alpha x$ lies in $K_{\mathcal{D}}$, since $c_i X = \gamma_i x \in \Sigma$. The normalized input X need not lie in Σ, which is why Proposition 5.2 was stated for arbitrary ternary inputs.

All coefficients constructed in the extraction are rational, and all exponents and visit times are ordinary integers. There are only countably many possible finite systems $\mathcal{D}$, including their unit anchors $\alpha_{\mathcal{D}}$. Hence

$$F_J \subseteq \bigcup_{\mathcal{D}} \alpha_{\mathcal{D}}^{-1} K_{\mathcal{D}}.$$

These are compact sets, and multiplication by a 3-adic unit is an isometry. By (23), each has upper box dimension at most $s_{r,u} < \delta$. This proves the covering statement.

Packing dimension is countably stable and is bounded above by upper box dimension on bounded sets. Therefore

$$\dim_{\mathrm{H}} F_J \le \dim_{\mathrm{P}} F_J \le \delta.$$

As $F_{k+1} \subseteq F_k$ and $\delta > 0$ was arbitrary, both limiting dimensions are zero. □

Remark 6.1 (Fixed finite-rank groups). The same proof applies to any fixed subgroup $G \le \mathbb{Q}^\times \cap \mathbb{Z}_3^\times$ of finite rank ρ. Replace distinct powers of two by distinct members $\gamma_i \in G$ and put $F_k(G) = \{x \in \mathbb{Z}_3 : \#\{\gamma \in G : \gamma x \in \Sigma\} \ge k\}$. In Lemma 2.2 the variable group has rank at most $\rho + d$, so replace the factor $d + 2$ in B_d by $\rho + d + 1$. The normalized solution tuples are still distinct: equality of their first coordinates gives $\gamma_i/\gamma_j = 3^{e_{t_0 i} - e_{t_0 j}}$; the left side is a unit, so the exponent difference is zero and $\gamma_i = \gamma_j$. All remaining arguments are unchanged. The coefficients stay rational, and the compact-cover data are countable. Thus for each $\delta > 0$ some finite $J = J(\delta, \rho)$ gives the same compact-cover conclusion for $F_J(G)$, and both finite-visit dimension limits are zero. The finite-rank hypothesis is essential to this argument; this remark does not concern arbitrary families of integer multipliers.

Corollary 6.2. *The infinite-visit exceptional set*

$$E_2 = \{x \in \mathbb{Z}_3 : 2^n x \in \Sigma \text{ for infinitely many } n \ge 0\}$$

has Hausdorff and packing dimension zero.

Proof. $E_2 \subseteq F_J$ for every J, so the result follows from Theorem 1.1. □

The finite-visit conclusion supplies uniformity in the number of visits, rather than a statement about a particular initial point. It does not imply that $1 \notin E_2$, that every nonzero orbit has finitely many visits, or that the exceptional set is countable. In particular no resolution of the pointwise Erdős problem is claimed. The effective constants supplied by repeated Ramsey numbers and the unit-equation bound are far too large for the result to serve as a practical search cutoff.

7 The real exceptional set and the proof dependencies

Define

$$\mathcal{C}_{\mathbb{R}} = \left\{ \sum_{j=-\infty}^{m} \varepsilon_j 3^j : m \in \mathbb{Z},\ \varepsilon_j \in \{0,1\} \right\} \subseteq [0,\infty),$$
$$E_{\mathbb{R}} = \{\lambda > 0 : 2^n \lambda \in \mathcal{C}_{\mathbb{R}} \text{ for infinitely many } n \geq 0\}.$$

The digit condition includes every fractional digit. This is the untruncated real exceptional set in Lagarias [1, §1.2], whose Conjecture 1.4 asks whether $\dim_{\mathrm{H}} E_{\mathbb{R}} = 0$. The corresponding statement is called Conjecture A in the arXiv manuscript [5]. It differs from the truncated set defined using only $\lfloor 2^n \lambda \rfloor$: Lagarias [1, Theorem 1.3] proved that the latter has Hausdorff dimension $\log_3 2$.

Theorem 7.1. *The untruncated real exceptional set satisfies*

$$\boxed{\dim_{\mathrm{H}} E_{\mathbb{R}} = \dim_{\mathrm{P}} E_{\mathbb{R}} = 0.}$$

More precisely, for every $\delta > 0$ there is a countable cover of $E_{\mathbb{R}}$ by compact subsets of $\mathbb{R}$ of upper box dimension at most δ.

The real proof uses the same finite causal-container theorem, Theorem 5.1, whose complete frozen proof from version 1.0.0 is reproduced in Appendix A. Its two-translate capacity estimate is Lemma 9.1 below; we reproduce its proof from [4, Lemma 2.1] as well. Thus neither finite input is left implicit. We give the full affine finite-word application, the transfer to real prefixes, and the dimension argument. We also restate and prove the arithmetic extraction for an arbitrary infinite set of visit times. Its external input is the finite-rank unit-equation theorem of Evertse, Schlickewei and Schmidt [3]. This infinite extraction uses a true visiting anchor; it is distinct from the uniform finite extraction with a virtual anchor in Theorem 2.1.

Throughout, logarithms and Shannon entropies are natural. Write

$$\Sigma_L = \left\{ \sum_{j=0}^{L-1} \varepsilon_j 3^j : \varepsilon_j \in \{0,1\} \right\} \subseteq \mathbb{Z}/3^L\mathbb{Z}.$$

For $z \in \mathbb{Z}_3$, let $[z]_m$ be its representative modulo 3^m in $[0, 3^m)$, and put

$$B_{j,q}(z) = \left[(z - [z]_j)/3^j \right]_q .$$

A rational number with denominator prime to 3 is interpreted in $\mathbb{Z}_3$ when residues or valuations are used. Its ordinary real value is used in floors and real intervals. This distinction is essential in the transfer below.

8 Templates from an infinite set of exponents

The following extraction is the version of [4, Lemma 3.1] that separates the arithmetic from the space in which the initial value lies.

Lemma 8.1. *Let $S \subseteq \mathbb{Z}_{\geq 0}$ be infinite, and fix integers $r \geq 3$, $K \geq 1$ and $q \geq 1$. There exist distinct indices $n_0, n_1, \ldots, n_K \in S$ and ordinary integer units $A_1, \ldots, A_{r-2} \in \mathbb{Z}_3^{\times}$ such that $c_i = 2^{n_i - n_0}$ has an exact expansion*

$$c_i = 1 + \sum_{t=1}^{r-2} A_t 3^{e_{ti}} + u_i 3^{E_i}, \qquad u_i \in \mathbb{Z}_3^{\times}, \tag{24}$$

where $0 < e_{1i} < \cdots < e_{r-2,i} < E_i$*, every successive gap, including the gap from* 0*, is at least* q*, and*

$$\min\{e_{1,i+1}, e_{2,i+1} - e_{1,i+1}, \ldots, E_{i+1} - e_{r-2,i+1}\} > E_i \qquad (1 \le i < K). \tag{25}$$

Proof. The distinct slopes $\alpha = 2^n$, $n \in S$, lie in the compact group $\mathbb{Z}_3^\times$. Choose a subsequence $\alpha_j \to \rho \in \mathbb{Z}_3^\times$. After discarding a possible equality, write

$$\rho^{-1}\alpha_j = 1 + v_{1j}3^{e_{1j}}, \qquad v_{1j} \in \mathbb{Z}_3^\times, \quad e_{1j} \to \infty.$$

Pass to a subsequence on which $v_{1j} \to a_1 \in \mathbb{Z}_3^\times$. Whenever $v_{1j} - a_1 \neq 0$, normalize this difference by its valuation and take a further convergent subsequence of its unit part. Repeating this procedure produces successive exponent gaps tending to infinity.

An infinite collection of exact stops after s terms would give infinitely many solutions of

$$\rho^{-1}\gamma - \sum_{t=1}^{s} a_t 3^{e_t} = 1, \qquad \gamma \in \langle 2 \rangle, \quad 0 < e_1 < \cdots < e_s.$$

The coordinate variables belong to the finite-rank multiplicative group $\langle 2 \rangle \times \langle 3 \rangle^s$. Every nonempty subsum on the left is nonzero: its terms have distinct 3-adic valuations $0, e_1, \ldots, e_s$, so the term of least valuation cannot cancel. The characteristic-zero finite-rank theorem of [3], applied over an algebraic closure of $\mathbb{Q}_3$, excludes infinitely many such nondegenerate solutions. Exact stops may therefore be discarded at each of the finitely many stages required here.

We obtain

$$\rho^{-1}\alpha_j = 1 + \sum_{t=1}^{r-2} a_t 3^{e_{tj}} + u_j^* 3^{E_j}, \qquad a_t, u_j^* \in \mathbb{Z}_3^\times,$$

with every successive gap tending to infinity. Choose K members, now indexed by i, so that the required lower gap bounds and (25) hold. Choose ordinary integer units A_t with

$$v_3(A_t - a_t) \ge \max_i (E_i - e_{ti} + q).$$

Finally choose another member $\alpha_0 = 2^{n_0}$ of the original subsequence, outside the selected finite set, with

$$v_3(\rho/\alpha_0 - 1) \ge \max_i (E_i + q).$$

In replacing α_i/ρ by α_i/α_0, and a_t by A_t, the total error in row i is divisible by 3^{E_i+q} in $\mathbb{Z}_3$. The normalized final remainder is consequently still a unit. This gives (24) with the same exponents and separation. □

Remark 8.2. The slopes c_i are positive rational 3-adic units, but need not be ordinary integers: the selected anchor n_0 can exceed some n_i. The proof used only the infinite set S, not an initial point in $\mathbb{Z}_3$. In particular, it applies to true visit times of a real point.

9 An affine finite-word estimate

Fix a template system as in Lemma 8.1. Choose $k \ge 1$ and $b \ge 1$ with $3^b > r - 1$, and use $q = b + k$ in the extraction. Set

$$R = \max_i E_i, \qquad Q = 3^{(r-1)q}, \qquad C = \binom{r}{2}, \qquad D = CQ^{r-3}.$$

For positive integer thresholds $T_{\mathrm{s}}, T_2, \dots, T_{r-1}$ define

$$\theta = Q\left[\frac{D\sum_{j=2}^{r-1} T_j + 1}{T_{\mathrm{s}}} + \frac{1}{T_2} + D\sum_{j=3}^{r-1} \frac{1+\sum_{l=2}^{j-1} T_l}{T_j}\right], \tag{26}$$

$$\epsilon_F = h_{\mathrm{bin}}(\theta) + \theta \log Q, \qquad h_{\mathrm{bin}}(t) = -t\log t - (1-t)\log(1-t), \tag{27}$$

$$\eta = \frac{Q(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j)}{K}, \qquad s = \frac{\epsilon_F}{\log 3} + \frac{1}{k} + \frac{2^{k+1}\log 2}{rk\log 3} + \eta. \tag{28}$$

We assume $\theta \le 1/2$.

The following estimate is reproduced from [4, Lemma 2.1]. It is uniform in both target translates.

Lemma 9.1 (Two-translate capacity)**.** *Suppose $A_1, \dots, A_r$ are nonempty subsets of $\mathbb{Z}/3^k\mathbb{Z}$ and*

$$T = A_1 + \cdots + A_r \subseteq (z_1 + \Sigma_k) \cup (z_2 + \Sigma_k).$$

Then

$$\sum_{i=1}^{r} \log|A_i| \le r\log 3 + 2^{k+1}\log 2 =: B_{r,k}. \tag{29}$$

Proof. Let $H = \mathrm{Stab}(T)$ and $|H| = 3^h$. A coset of H fixes the lowest $k-h$ digits. Its intersection with a translate of Σ_k has at most 2^h elements: after translating back, the highest h digits form a word of Σ_h. Since the nonempty H-periodic set T contains a whole coset, $3^h \le 2^{h+1}$. Thus $h \le 1$.

Set $n_i = |A_i + H|/|H|$. Kneser's addition theorem [6], applied in the quotient by H, gives

$$\sum_i (n_i - 1) \le |T|/|H| - 1 \le 2^{k+1}/|H| - 1.$$

Indeed the final sum in the quotient has trivial stabilizer; the stabilizer of any partial sum is contained in that of the final sum, so the two-summand aperiodic inequality may be applied successively. Since $n \le 2^{n-1}$ for positive integers n,

$$\sum_i \log|A_i| \le rh\log 3 + \sum_i (n_i - 1)\log 2 \le r\log 3 + 2^{k+1}\log 2.$$

□

Proposition 9.2. *For arbitrary fixed shifts $\beta_1, \dots, \beta_K \in \mathbb{Z}_3$ and $L \ge q$, let*

$$W_L(\boldsymbol{\beta}) = \{x \in \{0, \dots, 3^L - 1\} : [x]_L \in \Sigma_L,\ [c_i x + \beta_i]_L \in \Sigma_L\ (1 \le i \le K)\}.$$

Then, with $N = L - q + 1$,

$$\log|W_L(\boldsymbol{\beta})| \le Ns\log 3 + (R + b + k - 1)\log 3 \tag{30}$$

whenever the set is nonempty. Equivalently, including the empty case,

$$|W_L(\boldsymbol{\beta})| \le 3^{R+b+k-1}\, 3^{sL}. \tag{31}$$

Both bounds are uniform in the shifts. All locally admissible finite words are counted; an extension to an infinite admissible word is not required.

Proof. We give the affine application in detail because uniformity in the shifts is needed for the real transfer. View a finite input word as its ordinary representative $X \in [0, 3^L)$, extended by zero digits if needed, and set $Y_t = A_t X$ in $\mathbb{Z}_3$.

Exact causal arithmetic. For template i and source position j, put $v = j + E_i$. The participating windows are

$$B_0 = B_{j,q}(X), \quad B_t = B_{v-e_{ti},q}(Y_t) \ (1 \le t \le r-2), \quad B_{\mathrm{end}} = B_{v,q}(X).$$

Write $P = [X]_j$ and define

$$T = \beta_i + 3^{E_i} u_i P, \qquad t_v = [T]_v, \qquad D_j = (T - t_v)/3^v \in \mathbb{Z}_3.$$

The center D_j depends only on the fixed data and the input prefix $X_{<j}$. In the exact identity for $c_i X + \beta_i$, the lower parts are

$$[X]_v, \qquad 3^{e_{ti}}[Y_t]_{v-e_{ti}} \ (1 \le t \le r-2), \qquad t_v.$$

These are r nonnegative integers below 3^v, so their ordinary carry lies in $\{0, \dots, r-1\}$. Taking the next q digits gives the relation

$$u_i B_0 + \sum_{t=1}^{r-2} B_t + B_{\mathrm{end}} + D_j + e \in \Sigma_q \pmod{3^q} \qquad \text{for some } 0 \le e \le r-1. \tag{32}$$

Only windows contained in the first L output digits will be used. Thus this relation is valid for every word in $W_L(\boldsymbol{\beta})$.

Low digits and capacity. On any Cartesian product of windows satisfying (32), fix the lowest b digits of each argument. The low residue of their sum, including D_j, is fixed. If $y \in \Sigma_q$ is an allowed output, then

$$y_{\mathrm{low}} - e \in [-(r-1), 3^b - 1].$$

Since $3^b > r-1$, a fixed residue modulo 3^b has at most two representatives in this interval. Dividing the corresponding identities by 3^b shows that the sum of the high k-digit factors lies in at most two translates of Σ_k. The first high factor is multiplied by the unit u_i, an affine bijection after its low digits are fixed. We use Lemma 9.1: for nonempty $H_1, \dots, H_r \subseteq \mathbb{Z}/3^k\mathbb{Z}$,

$$H_1 + \cdots + H_r \subseteq (z_1 + \Sigma_k) \cup (z_2 + \Sigma_k) \quad \Longrightarrow \quad \sum_{\nu=1}^{r} \log |H_\nu| \le r \log 3 + 2^{k+1} \log 2 =: B_{r,k}. \tag{33}$$

The proof above derives this estimate from Kneser's theorem; its constants do not depend on either translate.

The finite causal-container input. At each position $j \in \{0, \dots, N-1\}$ take the joint letter

$$Z_j = (B_{j,q}(X), B_{j,q}(Y_1), \dots, B_{j,q}(Y_{r-2}))$$

in an alphabet of size Q. The ordered positional edges are

$$(j, j + E_i - e_{r-2,i}, \dots, j + E_i - e_{1i}, j + E_i)$$

whenever all positions belong to $\{0, \dots, N-1\}$. Their relations select the indicated coordinates in (32). By (25), future position sets of distinct edges from the same source are disjoint. Every pair of positions belongs to at most $C = \binom{r}{2}$ edges: the distance belongs to at most one template, and that template has at most $\binom{r}{2}$ role pairs.

Theorem 5.1, with its complete proof in Appendix A, therefore supplies a deterministic fingerprint F and alphabets $\mathcal{A}_j$, each determined by $(X_{<j}, F)$ and containing the actual joint letter. It gives

$$H(F) \le N\epsilon_F, \qquad E_{\mathrm{bad}} \le Q\left(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j\right) N. \tag{34}$$

Except for at most E_{bad} edges, the *full Cartesian product* of the assigned alphabets satisfies its relation at the actual source prefix. This full-product conclusion, rather than compatibility along one input, is the property used next.

Conditional supports and entropy. Take X uniformly distributed on the nonempty set $W_L(\boldsymbol{\beta})$. For each realized word put

$$C_j = \mathrm{supp}(X_{[j+b,j+b+k)} \mid X_{<j+b}, F), \qquad w_j = \log|C_j| \le k\log 3.$$

Fixing $X_{<j+b}$ fixes the low b digits of all joint windows at j. Multiplication by any A_t induces an affine bijection on the next k digits. Each required high-coordinate projection of the conditional joint support consequently has exactly $|C_j|$ elements. This joint support is contained in $\mathcal{A}_j$, since $\mathcal{A}_j$ is determined by the shorter prefix $X_{<j}$ and the same fingerprint.

On a good edge these projections may be selected independently, even though the prefixes defining them differ at the different positions. The full-product property applies to every such selection; no common new input realizing all selections is asserted. The low-digit reduction and (33) imply, pointwise on that edge,

$$\sum_{\nu=0}^{r-1} w_{j_\nu} \le B_{r,k}.$$

On a bad edge use $rk\log 3$. Template i has $\max(N - E_i, 0)$ edges, and each of its r roles misses $\min(E_i, N)$ positions. Accounting for these boundary incidences gives

$$rK\sum_{j=0}^{N-1} w_j \le B_{r,k}KN + rk\log 3\, E_{\mathrm{bad}} + rk\log 3\, KR,$$

and hence

$$\sum_j w_j \le \frac{NB_{r,k}}{r} + k\log 3\left(\frac{E_{\mathrm{bad}}}{K} + R\right). \tag{35}$$

For $h_t = H(X_t \mid X_{<t}, F)$ the entropy chain rule yields

$$\sum_{t=j+b}^{j+b+k-1} h_t \le \mathbb{E}w_j.$$

The multiplicities of the h_t in this sum are at most k and total kN. Since $h_t \le \log 3$, their deficit from kL is at most $k(q-1)\log 3$ in entropy units. Therefore

$$H(X \mid F) \le k^{-1}\sum_j \mathbb{E}w_j + (q-1)\log 3.$$

Combining this with (34)–(35), and using $H(X) = H(F) + H(X \mid F) = \log|W_L(\boldsymbol{\beta})|$, proves (30). Dropping its nonpositive contribution $-(q-1)s\log 3$ gives (31). All bounds are independent of the shifts, as required. □

10 Transfer from real truncations to finite words

The next lemma is the additional step connecting the two number systems. It requires a bound for all finite words, not merely for prefixes that extend to infinite admissible 3-adic inputs.

Lemma 10.1 (Finite-word transfer)**.** *Let $c_i = p_i/q_i > 0$ be rational numbers in lowest terms, with $3 \nmid p_i q_i$, for $1 \le i \le K$. Suppose that, for some constants $C_0 > 0$, $s \ge 0$ and L_0, their affine word sets satisfy*

$$|W_L(\boldsymbol{\beta})| \le C_0\, 3^{sL} \quad (L \ge L_0)$$

for all $\boldsymbol{\beta} \in \mathbb{Z}_3^K$. For each integer $M \ge 1$, put

$$K_{\mathbb{R}}(\boldsymbol{c}; M) = \{X \in [0, M] : X \in \mathcal{C}_{\mathbb{R}},\ c_i X \in \mathcal{C}_{\mathbb{R}}\ (1 \le i \le K)\}.$$

This set is compact and has upper box dimension at most s.

Proof. On a bounded interval, there are only finitely many possible integer digit prefixes in $\mathcal{C}_{\mathbb{R}}$. Each has a compact set of fractional tails, the continuous image of the compact product $\{0,1\}^{\mathbb{N}}$. It follows that $\mathcal{C}_{\mathbb{R}}$ is closed on bounded intervals, and the stated finite intersection is compact.

Fix $X \in K_{\mathbb{R}}(\boldsymbol{c}; M)$ and $L \ge 1$. Define

$$T_L = \lfloor 3^L X \rfloor, \qquad T_{i,L} = \lfloor 3^L c_i X \rfloor, \qquad U_L = T_L \bmod 3^L \in \{0, \dots, 3^L - 1\}.$$

The full real digit conditions imply $U_L \in \Sigma_L$ and $T_{i,L} \bmod 3^L \in \Sigma_L$. Indeed, after multiplication by 3^L, the remaining fractional 0/1 tail lies in $[0, 1/2]$, so taking the floor gives exactly the allowed integer prefix. This also handles terminating expansions. Set

$$\beta_{i,L} = T_{i,L} - c_i T_L.$$

If $\tau = \{3^L X\}$ and $\tau_i = \{3^L c_i X\}$, then exactly

$$\beta_{i,L} = c_i \tau - \tau_i \in q_i^{-1}\mathbb{Z} \cap (-1, p_i/q_i) =: \mathcal{B}_i. \tag{36}$$

This is a fixed finite set of cardinality $p_i + q_i - 1$. Every element belongs to $\mathbb{Z}_3$ because $3 \nmid q_i$.

Now $T_L - U_L$ is an integer multiple of 3^L, so in $\mathbb{Z}_3$

$$c_i U_L + \beta_{i,L} = T_{i,L} - c_i(T_L - U_L) \equiv T_{i,L} \pmod{3^L}.$$

Consequently

$$U_L \in W_L(\beta_{1,L}, \dots, \beta_{K,L}). \tag{37}$$

The vector of shifts may change with X and with L. We take the finite union over $\prod_i \mathcal{B}_i$ at each depth; no consistency of a single shift vector across depths is needed.

For a fixed integer part of X, the value U_L determines a real half-open cell of length 3^{-L}. The first L fractional digits form the ordinary base-three integer U_L in their usual written order; the associated 3-adic word reads these same digits from low to high. Both orders preserve the condition that every digit is 0 or 1. There are at most $M + 1$ integer parts. Thus the number of occupied real cells of this length is at most

$$(M+1)\left(\prod_{i=1}^{K}(p_i + q_i - 1)\right) C_0\, 3^{sL}, \qquad L \ge L_0. \tag{38}$$

Closed intervals give a cover with the same bound up to a fixed factor. The factors preceding 3^{sL} are independent of L, so the upper box dimension is at most s. The discrete scales 3^{-L} suffice, since consecutive scales have bounded ratio. □

Remark 10.2. The variability of the shifts is real, even for fully admissible numbers. For $X = 1/2$ and $c = 1/4$, both X and $cX = 1/8$ belong to $\mathcal{C}_{\mathbb{R}}$, while (36) equals $-1/4$ for odd L and 0 for even L. Applying the container theorem after fixing each vector in the finite union is what preserves prefix dependence. The proof does not feed an unread real tail into a causal relation.

11 Countable compact covers and the dimension conclusion

Proof of Theorem 7.1. Fix $\delta > 0$. Choose first k sufficiently large, and then r so large that

$$\frac{1}{k} + \frac{2^{k+1} \log 2}{rk \log 3} < \delta.$$

Leave a positive margin in this inequality. Choose b with $3^b > r - 1$, and set $q = b + k$, Q, C, D as in Section 9. For integers $u \to \infty$ use the thresholds and number of templates

$$T_j = u^{2^{j-1}-1} \quad (2 \le j \le r-1), \qquad T_{\mathrm{s}} = u^{2^{r-1}-1}, \qquad K = u^{2^r}.$$

Every ratio in (26) tends to zero. Thus $\theta, \epsilon_F, \eta \to 0$, and we can fix one finite u with $\theta \le 1/2$ and $s \le \delta$ in (28). All these parameters are now fixed.

For $\lambda \in E_{\mathbb{R}}$ the true visit set

$$S(\lambda) = \{n \ge 0 : 2^n \lambda \in \mathcal{C}_{\mathbb{R}}\}$$

is infinite. Apply Lemma 8.1 to this set, obtaining true indices $n_0, n_1, \ldots, n_K$ and finite template data. Then $X = 2^{n_0}\lambda$ and $c_i X = 2^{n_i}\lambda$ all belong to $\mathcal{C}_{\mathbb{R}}$. For some positive integer M, therefore,

$$X \in K_{\mathbb{R}}(\boldsymbol{c}; M).$$

By Proposition 9.2 and Lemma 10.1, this compact set has upper box dimension at most $s \le \delta$.

Let $\mathcal{D}$ range over the finite data consisting of the visit indices, the integer exponents and the ordinary integer coefficients in Lemma 8.1. There are only countably many such data. We have proved the cover

$$E_{\mathbb{R}} \subseteq \bigcup_{\mathcal{D}} \bigcup_{M \ge 1} 2^{-n_0} K_{\mathbb{R}}(\boldsymbol{c}(\mathcal{D}); M). \tag{39}$$

No 3-adic accumulation point or limiting coefficient indexes this union. Each set is compact, and multiplication by 2^{-n_0} is a real similarity, so its upper box dimension is at most δ.

The span R and the rational heights p_i, q_i may be unbounded across the cover. This causes no problem: in (38) the depth L tends to infinity for one fixed finite system at a time. All of those quantities affect only its multiplicative counting constant. Countable stability of packing dimension and the usual inequalities $\dim_{\mathrm{H}} \le \dim_{\mathrm{P}} \le \overline{\dim}_{\mathrm{B}}$ on each compact component now imply

$$\dim_{\mathrm{H}} E_{\mathbb{R}} \le \dim_{\mathrm{P}} E_{\mathbb{R}} \le \delta.$$

Since $\delta > 0$ was arbitrary, both dimensions are zero. □

The real argument uses infinitely many visits in its extraction step; no uniform finite-visit conclusion for the real sets is asserted here. The 3-adic finite-visit conclusion of Theorem 1.1 was proved separately using the uniform finite extraction in Theorem 2.1. A zero-dimensional exceptional set need not be countable or empty. In particular, Theorem 7.1 does not settle whether $1 \in E_{\mathbb{R}}$, the remaining pointwise question associated with the powers of two.

The positive-dimensional truncated exceptional set does not contradict the transfer: a condition only on $\lfloor 2^n \lambda \rfloor$ gives no restriction on arbitrarily long fractional prefixes. In that setting the membership assertion (37) is unavailable.

12 Exact checks for the real transfer

The real proof consists of the arguments above, the reproduced finite causal-container proof in Appendix A, and the published unit-equation and sumset theorems [3, 6]. The finite inputs originating in version 1.0.0 are explicitly identified. The accompanying exact-arithmetic program checks the new finite-word bridge using rational numbers, including terminating and periodic ternary expansions. It verifies the bounds on the shifts, reduction with rational unit denominators, and membership of the resulting finite words. The current finite test set contains 1845 rational inputs and 49 slopes, with 632835 checked identities, including 12215 instances with fully admissible input and output expansions. Negative and nonintegral shifts are included, as is the alternating-shift example in Remark 10.2.

These computations detect errors in the finite identities; they do not prove the dimension theorem or replace the causal-container argument. ChatGPT was used in developing and editing the manuscript, implementing the exact checks, and conducting internal AI-assisted reviews from several mathematical perspectives. Such reviews do not constitute external peer review or proof-assistant verification. Authorship of this manuscript is attributed solely to Timofei Iuzhakov.

Dependency and verification record

The only imported nonstandard finite result is Theorem 5.1, from Appendix A of the exact release v1.0.0 of [4], commit

`81c7f5ae45c3ce507173b82baca48fdfeb52c6ef`

Its complete proof is included unchanged in Appendix A. The SHA-256 digest of the appendix source `causal_container.tex` is

`1ae15961cd3bc705e7313c9e56de34e5127`
`7cffaa7cc3a58c6e4c9cb6e672750`

with the two displayed lines concatenated. This identifies the exact imported proof on which the finite entropy estimate depends. The unit-equation theorem is the published result [3]; finite Ramsey theory is used only in its ordinary finite form. The real argument also uses Kneser's addition theorem [6], through the full proof of Lemma 9.1 reproduced from [4, Lemma 2.1]. The statements and proofs of the uniform finite extraction and of the real infinite-set extraction are kept separate. The present version does not assert a real finite-visit nesting theorem, a finite-visit Assouad theorem, or an arbitrary-target generalization.

The arithmetic extraction and the entropy transfer were subjected to independent internal AI-assisted reviews. Exact finite checks were also used to test determinant identities and low-dimensional cases. These checks do not replace the arguments for arbitrary d, all Ramsey refinements, or the causal-container theorem. AI assistance was used in developing proofs, writing the manuscript, and implementing checks. This is not external peer review or proof-assistant verification. Authorship is attributed solely to Timofei Iuzhakov.

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A The finite causal-container theorem and its proof

The following source is reproduced unchanged from Appendix A of [4], release v1.0.0. Its symbols are local to this appendix.

A.1 Exact theorem

Let $r \geq 3$, let $V_0 = \{0, \dots, N-1\}$, let Ω be a finite alphabet of size Q, and let $\mathcal{W} \subseteq \mathcal{X}^L$ be nonempty, where $\mathcal{X}$ is any finite input alphabet. For every $i \in V_0$, a word x determines a letter $Z_i(x) \in \Omega$. No locality hypothesis on the maps Z_i is needed for this finite theorem.

Let $\mathcal{E}$ be a fixed family of ordered positional edges

$$e = (s = i_0 < i_1 < \cdots < i_{r-1}).$$

For each edge there is a relation $\mathcal{R}_e(x_{<s}) \subseteq \Omega^r$, depending only on the indicated input prefix, and every $x \in \mathcal{W}$ satisfies

$$(Z_{i_0}(x), \dots, Z_{i_{r-1}}(x)) \in \mathcal{R}_e(x_{<s}).$$

Assume:

1. for a fixed source s, the future position sets $e \setminus \{s\}$ of distinct outgoing edges are disjoint;
2. every unordered pair of positions belongs to at most C positional edges.

Fix positive integer thresholds $T_{\mathrm{s}}, T_2, \dots, T_{r-1}$, and put

$$D = CQ^{r-3},$$

$$\theta = Q\left[\frac{D\sum_{j=2}^{r-1} T_j + 1}{T_{\mathrm{s}}} + \frac{1}{T_2} + D\sum_{j=3}^{r-1} \frac{1 + \sum_{l=2}^{j-1} T_l}{T_j}\right].$$

Assume $\theta \leq 1/2$. Then each word can be assigned a set $F(x)$ of at most θN pairs $(i, Z_i(x))$, with at most one pair for each position, and nonempty sets $A_i(x) \subseteq \Omega$, such that:

- $Z_i(x) \in A_i(x)$;
- $A_i(x)$ is determined by $(x_{<i}, F(x))$;
- except for at most

$$Q\left(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j\right) N$$

edges, the full product of the assigned alphabets is compatible:

$$\prod_{\nu=0}^{r-1} A_{i_\nu}(x) \subseteq \mathcal{R}_e(x_{<i_0});$$

- for every distribution on $\mathcal{W}$,

$$H(F) \le N\big[h(\theta) + \theta \log Q\big], \qquad h(t) = -t\log t - (1-t)\log(1-t).$$

The window parameter q in the application only sets $N = L-q+1$; the abstract theorem itself does not involve windows.

A.2 Literals, truth, and complete origins

Fix one realized word x for the construction and counting arguments, and write $z_i = Z_i(x)$. A **literal** is a pair (i, a) with $a \in \Omega$. A literal is true if $a = z_i$. There are QN literals.

A **complete origin** consists of:

- one positional edge $e = (s, i_1, \ldots, i_{r-1})$;
- one assignment $(a_1, \ldots, a_{r-1})$ of letters to its future positions;
- the condition

$$(z_s, a_1, \ldots, a_{r-1}) \notin \mathcal{R}_e(x_{<s}).$$

The source argument is the one fixed true source letter. An origin is therefore a forbidden future set of $r-1$ literals. It always contains a false literal: otherwise the realized word would violate its original relation.

A projection of an origin removes some true literals whose positions have been explicitly recorded. An origin becomes irrelevant when any of its required literals is excluded from a palette. An origin may also be discarded at the chronological retirement of an unrecorded position; this latter loss will be charged to the bad-edge budget.

A.2.1 Pair multiplicity lemma

For any two future literals $v = (i, a)$, $w = (j, c)$ at distinct positions, at most $D = CQ^{r-3}$ complete origins contain both.

Proof. There are at most C original positional edges containing the two positions. In any origin whose future set contains both literals, its source is earlier than both and its source letter is already fixed to the true letter. The two prescribed future letters leave only $r-3$ future letters unspecified. There are at most Q^{r-3} possibilities. Sum over the at most C edges. □

This is a bound on distinct full origins over the entire execution for this word. It is not a bound on the number of historical projection shapes. The distinction is essential: an origin can appear successively at several ranks, but its complete label is unchanged.

One may define the universe of all complete origins at the outset for purposes of counting, even though the actual algorithm enumerates an origin only when its source is processed. Thus an origin selected just before synchronous palette deletions can be charged to those deletions even if its queue job has not yet been materialized.

A.3 State and invariants

Each position i has a palette S_i, initially Ω. Palettes only shrink, and the true letter is always retained.

A position is **recorded** when its true letter is put in the fingerprint. Its palette then becomes the singleton $\{z_i\}$. A recorded true literal remains in that palette but is inactive in all graphs.

An unprocessed, unrecorded literal in its current palette is active. At the chronological retirement of a position, all its remaining active literals become inactive permanently. Inactivity and palette exclusion are different operations.

For each $2 \le j \le r-1$, maintain a simple j-uniform hypergraph J_j on active literals. Every physical hyperedge carries one or more complete-origin labels. All labels carried by a physical edge have that edge as their current projection.

Also maintain a finite queue of jobs. A job contains a projected set of literals, the full-origin labels it owns, and a frozen rank bound. Its actual remaining arity may fall below that bound when recorded literals are substituted or excluded literals are noticed. The bound never increases along an origin's history.

We maintain:

I1. Truth preservation. Every palette contains its true letter.

I2. Unique ownership. Each live origin has exactly one current owner: one physical edge or one queued job. Equal physical edges merge their label sets. Origins never branch.

I3. Sound projection. A live projection is obtained from its complete origin only by removing recorded true literals. No unrecorded true value is silently substituted.

I4. No false resurrection. A literal excluded from its palette never returns. An inactive literal never becomes active again.

I5. Degree control. Immediately before every physical insertion, all active degrees in each J_j are strictly less than T_j. At the instant any active literal is removed, its degree in J_j is at most T_j.

I6. Source barriers. Before the chronological retirement of a position, and before the next source is processed, every queue is empty.

All choices—edge order, literal order, witness order, and queue tie breaking—are fixed deterministically from the available state. The full realized word is used only to answer an explicitly posed equality query and to select the heavy or light actual source branch. Replay will eliminate the latter dependence.

A.4 Primitive operations

A.4.1 Excluding a false literal

Delete the literal from its palette and from the active graphs. Every active graph origin requiring it becomes irrelevant. A queued origin requiring it is likewise irrelevant, even if its job discovers this only later. Record the time of the actual palette exclusion for accounting; a later inspection of the job is not a second deletion event.

A negative equality query is one way to exclude a false literal. Unit propagation from a valid forbidden projection is another.

A.4.2 Recording a true literal

For an active literal $v = (i, z_i)$:

1. first snapshot and take ownership of all physical edges incident to v, in every rank, with their origin labels;

2. add (i, z_i) to the fingerprint; exclude all other palette letters at position i; deactivate v, preserving its singleton palette;
3. for every snapshotted rank-two edge $\{v, w\}$, exclude w, before inserting any new physical edge;
4. for every other snapshotted edge B, put $B \setminus \{v\}$, with its origin labels, into a queue with frozen rank $|B| - 1$.

The order can be implemented without ambiguity by first removing all snapshotted physical edges from the graph and giving their labels to their projected jobs or unit consequences. Exclusions can make some of those projected jobs irrelevant before they are later inspected.

Every rank-two neighbor is false: if it were true, an origin of the edge, together with the previously substituted true literals, would make the original forbidden assignment entirely true. Thus I1 is preserved.

Edges contain at most one literal from any position, because they are projections of positional edges. In particular, the other letters excluded at position i do not form second ends of the saved edges incident to v.

A.4.3 Processing a job

If a required letter has been excluded from its palette, discard the job's origins. Otherwise substitute every recorded true literal in it. A letter at a recorded position that does not match the recorded value was already excluded and falls under the first case.

There cannot be zero remaining literals, because then a forbidden origin would be entirely true. With one remaining literal, exclude that false literal. With $j \geq 2$ remaining literals, insert their set into J_j, or merge the origin labels if that physical edge already exists.

A merge does not increase any physical degree and is not a physical insertion.

A.4.4 Degree-triggered queries

Suppose a new physical edge of rank j is inserted. Before insertion every degree is below T_j, so only its ends can reach T_j, and none can exceed it. If an end reaches T_j, choose one such end v and query whether it is true.

A negative response excludes v. A positive response records v. In either case the newly inserted edge disappears immediately. Every other end that had just reached the threshold loses this same new edge and returns below threshold. At that moment other ranks have undergone only removals.

The only remaining operations that can increase degrees are later physical insertions, each handled by the same rule. This proves I5 inductively, once the scheduler below ensures that new insertions are processed one at a time.

A.5 An explicit packet scheduler

A positive query that is triggered in rank j opens its rank-j packet **before** the snapshot and recording operation. Thus the packet includes synchronous exclusion of the other letters at the recorded position and all saved rank-two neighbors.

After recording, drain all pending jobs whose frozen rank is strictly less than j. Each such job has actual arity at most its frozen rank, so any positive query it triggers has rank strictly less than j. That query opens a nested packet of strictly smaller rank. On return, continue draining eligible jobs until none remains. Then close the rank-j packet.

Jobs of frozen rank at least j are deferred while this packet is open. A parent packet can process them later if their frozen rank is below the parent's bound. An external drain with bound $+\infty$ processes all jobs.

This scheduler yields three facts used below:

1. nested positive packets have strictly decreasing ranks;
2. two packets of the same rank have disjoint time intervals;
3. an origin that started as the projection of a saved rank-j edge can never be stranded in the rank-j packet's deferred queue: its initial projection has frozen rank $j-1$, and every subsequent projection along that origin's chain has still smaller frozen rank.

A job that came from an unrelated larger edge can have actual arity below j while retaining frozen rank at least j; deferring it is harmless. The third fact concerns the selected own origins of the packet, whose frozen ranks are already below j.

There is no chronological retirement inside a packet. Chronological retirement happens only after the external drain has finished.

Termination is finite. Only finitely many original forbidden assignments can be installed. Each origin has one chain, and every transfer created by recording strictly reduces its number of required literals. A job is processed once unless it is replaced by such a strict projection; it is otherwise discarded or attached to one physical edge. No origin is duplicated. There are therefore finitely many insertions, transfers, and query events.

A.6 Processing sources

Process sources $i = 0, \dots, N-1$ in order. Queues are empty at entry. The relations whose source is i are known from $x_{<i}$.

If i is already recorded, assign $A_i = \{z_i\}$, enumerate every forbidden future tuple with source letter z_i in the current future palettes, and externally drain all resulting jobs.

If i is not recorded, call a candidate $a \in S_i$ **heavy** if at least T_{s} outgoing positional edges admit a forbidden future tuple in the current future palettes when their first letter is a. Otherwise it is light.

If the true letter is heavy:

- snapshot the current palettes and the full list of forbidden future tuples for the true source letter;
- begin the source accounting interval before any recording or deletion;
- record the source, assign $A_i = \{z_i\}$, and enqueue the snapshotted list;
- externally drain every queue.

If the true letter is light, assign A_i to be all light letters of S_i. This nonempty set contains the true letter. This branch creates no new constraints, no new jobs, and no queries.

Finally, with all queues empty, retire the remaining active literals at position i. Delete all physical edges incident to them, discarding their origins without substituting the unrecorded actual value. Do not change the assigned A_i afterward.

The future positions of any newly installed origin are later than its source. Thus a newly installed origin never contains an already retired position. At a chronological retirement, every earlier queued job has been processed; all still-live origins requiring that position are on its active incident edges and are discarded then. Consequently no later event refers to a retired position. This also proves that future palette changes never alter an already assigned alphabet.

A.7 Physical insertion budget

Let I_j be the number of physical insertions in rank j. Every inserted edge is eventually removed, because all its positions are ultimately retired or recorded.

Charge an insertion to the first removal of one of its active endpoints. Each literal is removed from active status once, and its degree at that event is at most T_j. There are QN literals. Therefore

$$I_j \le T_j QN. \tag{7.1}$$

A physical edge with a given literal set cannot disappear and later be reinserted with that same literal set: its first disappearance removes at least one of those active literals permanently. This is also a direct reason why it is legitimate to speak of the unique insertion event of a surviving physical edge. Alternatively, charging by insertion occurrence gives the same estimate.

A.8 Positive-query budgets: the central charge maps

Let H_j be the number of positive rank-j queries, and let H_{s} be the number of newly recorded heavy sources.

A.8.1 Rank two

At a positive rank-two query, the queried literal has exactly T_2 distinct physical neighbors. All are active and present in their palettes at that instant. Recording removes these T_2 distinct false neighbors before any new insertion.

Each literal can be excluded from its palette at most once. Hence

$$H_2 T_2 \le QN. \tag{8.1}$$

Neighbors may belong to the same position but be different letters; they are still different literals, which is exactly what the QN budget counts.

A.8.2 Rank $j \ge 3$: selection

At a positive rank-j query with true pivot v, its snapshot has exactly T_j incident physical edges. Select one complete-origin label from each. These labels are distinct by unique ownership.

Immediately after substituting v, every selected origin has a projection of rank $j-1$. Such an origin is never selected again by a different rank-j packet, because its single chain can never return to rank j. Thus there are exactly $H_j T_j$ distinct rank-j selected-origin instances over all such packets.

A.8.3 Rank $j \ge 3$: exhaustive exit alternatives

For each selected origin, at the closure of its packet exactly one of the following accounting alternatives applies:

- it has lost a required palette letter during this packet;
- it is carried by an active physical edge of rank $l < j$.

Here is why this list is exhaustive. No chronological retirement occurs within the packet. The origin started alive and its actual projection has rank $j-1$. Every subsequent job on its chain has frozen rank below j, so cannot remain in the deferred queue when the packet closes. If a recorded end is substituted, the rank decreases further. A zero-end origin is impossible by soundness. A one-end origin causes a palette exclusion. If it is neither irrelevant nor still queued, unique ownership places it on an active physical edge, necessarily of smaller rank.

A.8.4 Charging actual palette losses

For the first alternative, charge the selected origin to its first required palette exclusion during the packet. That excluded literal w and the packet pivot v both belong to the origin's full future assignment. They are distinct: the true pivot is recorded and retained, not excluded.

For one fixed packet and one fixed exclusion event w, the pair lemma bounds these selected origins by D. Packets of the same rank are time-disjoint, so an actual exclusion event belongs to at most one rank-j packet. There are at most QN exclusion events overall. The total loss charge is therefore at most DQN.

This includes exclusions at packet opening. An origin whose queued projection is inspected later is charged to the earlier actual exclusion event within the same packet. Nothing is charged to a merely administrative late inspection.

A.8.5 Charging surviving lower edges

For the second alternative, charge the selected origin to the unique insertion event of its final active physical edge B of rank $l < j$.

Fix any pair of literals in B. Every complete origin charged to this edge contains that pair. Across all rank-j packets, each complete origin is selected at most once at rank j, and at most D complete origins contain the pair. The physical insertion consequently pays for at most D selected origins, even if some origins merge into an edge created before their packets began.

This is the precise resolution of the "old lower edge receives new projections" concern. The budget counts its full-origin labels over time, not its creation time relative to the paying packet.

Combining both charge classes gives

$$H_j T_j \le D\left(QN + \sum_{l=2}^{j-1} I_l\right), \qquad 3 \le j \le r-1. \tag{8.2}$$

A.9 Heavy-source budget

At a newly recorded heavy source choose T_{s} different outgoing positional edges witnessing heaviness, and one forbidden future tuple from each, in the palettes snapshotted before recording.

Their future position sets are pairwise disjoint. These tuples are alive at the start of the source accounting interval. At the end of the external drain, a selected origin either has lost a palette letter during this interval or is on an active physical edge. There are no queued jobs, no chronological retirements during the interval, and no completely true forbidden tuple.

An actual excluded literal belongs to at most one selected future tuple for this source, by future-position disjointness. Source intervals are successive and disjoint. Thus palette-loss charges total at most QN.

For surviving-edge charges, any fixed pair of the active edge belongs to every paying complete origin; a source is processed only once and one origin was chosen for each selected outgoing edge. The pair lemma bounds charges to a physical insertion by D, over all source intervals.

Therefore

$$H_{\mathrm{s}} T_{\mathrm{s}} \le QN + D\sum_{j=2}^{r-1} I_j. \tag{9.1}$$

It is legitimate for a deletion or insertion to participate in the separate upper bounds for different ranks and for heavy sources. These are different estimates for different record counts; each estimate has its own valid charge map. The proof never asserts a single shared global budget across all these independently bounded classes.

A.10 Fingerprint size

A position is recorded only once. Every recording is either a new heavy-source recording or a positive query recording. Thus

$$|F| = H_{\mathrm{s}} + \sum_{j=2}^{r-1} H_j.$$

Using (7.1), (8.1), (8.2), and (9.1) gives

$$H_{\mathrm{s}} \le QN \frac{D \sum_{j=2}^{r-1} T_j + 1}{T_{\mathrm{s}}},$$

$$H_2 \le QN/T_2,$$

$$H_j \le DQN \frac{1 + \sum_{l=2}^{j-1} T_l}{T_j} \quad (j \ge 3).$$

Their sum is exactly θN, with the stated constant. No hidden dependence on N, the relations, or the realized word appears.

A.11 Bad-edge budget

Call an edge bad for this word if its assigned full Cartesian product fails to be contained in the relation frozen at its actual source prefix.

A.11.1 Light sources

For a light source, each of its at most Q assigned source candidates has fewer than T_{s} outgoing edges with a forbidden future tuple in the then-current future palettes. Later future alphabets are subsets of those palettes. Consequently the source has fewer than QT_{s} bad outgoing edges.

The total light-source contribution is at most $QT_{\mathrm{s}}N$.

A.11.2 Recorded sources

For a bad edge with recorded source, choose one forbidden tuple from the final assigned alphabets. Its source letter is the actual recorded letter. Its future letters were all present when its source was processed, because palettes shrink and assigned alphabets are never enlarged after retirement. Hence this full origin was enumerated or was already rendered irrelevant only by a palette event that would also exclude a final alphabet letter.

The latter cannot happen: every chosen future letter belongs to its assigned final alphabet, so none was ever excluded from its palette. Recording one of its future positions can only substitute the matching true letter; recording a different letter would exclude the chosen one, again contradicting final membership.

Follow the origin's unique chain. It cannot end in a unit exclusion or an empty projection, and it cannot disappear through a false-letter exclusion. At the end of the complete execution

no active literal remains. Because every chronological retirement has an empty queue, the origin must first be discarded on an active physical edge at a chronological retirement.

Charge this bad positional edge to that physical edge's insertion event, using the selected complete origin as the invoice. Different bad positional edges have different complete origins. Every origin charged to a fixed physical edge contains any chosen pair of its literals. The pair lemma therefore bounds its charges by D.

The recorded-source contribution is at most $D\sum_j I_j$. Together with (7.1),

$$E_{\mathrm{bad}} \le QT_{\mathrm{s}}N + D\sum_{j=2}^{r-1} I_j \le Q\left(T_{\mathrm{s}} + D\sum_{j=2}^{r-1} T_j\right)N.$$

This bound explicitly pays for chronological deletion without substituting an unrecorded actual value. Omitting that payment would leave a real gap; it is included in the bound here.

A.12 Exact replay from the final fingerprint

Fix a realized pair $(x_{<i}, F(x))$. We show that the entire execution through assignment of A_i can be reconstructed from this pair.

At any equality query on an active literal (k, a), the answer is positive if and only if $(k, a) \in F(x)$:

- every positive answer immediately records the pair;
- every recorded pair is true;
- a true literal cannot be excluded or change its value later.

The fingerprint is used only to answer the queries actually produced by the deterministic algorithm. It is not used to insert all future recorded letters into the state in advance. Absence of an unqueried pair says nothing by itself about that letter.

Process earlier sources by induction. Their relations are known because their source prefixes are restrictions of $x_{<i}$. Each queue action and literal query is then determined by the reconstructed state and the fingerprint oracle.

The only remaining branch is a source $s \le i$ that has not already been recorded when its turn arrives. At that point all queues are empty.

- If $s \notin \operatorname{dom} F$, its actual source letter cannot be heavy, because that would record it. Its actual letter is light. The algorithm computes A_s as all light letters from the known state and creates no events.
- If $s \in \operatorname{dom} F$, its true letter is given by its fingerprint pair. If that letter were light, the light branch would create no events; the source would immediately retire, after which it could never be recorded. This contradicts membership in the final fingerprint. Therefore the actual letter is heavy and the heavy branch is determined.

An already recorded source uses its known recorded value. This completes the induction. The algorithm determines A_i from $(x_{<i}, F)$ exactly as claimed.

Consequently, for any probability distribution on $\mathcal{W}$, the assigned A_i contains the entire conditional support

$$\operatorname{supp}(Z_i \mid x_{<i}, F),$$

on every conditioning atom of positive probability. This is stronger than merely containing the one actual letter on one execution.

A.13 Fingerprint entropy

There are at most

$$\sum_{a=0}^{\lfloor \theta N \rfloor} \binom{N}{a} Q^a$$

possible fingerprints. Since $\theta \le 1/2$,

$$\sum_{a \le \lfloor \theta N \rfloor} \binom{N}{a} \le \exp\{N h(\theta)\}, \qquad Q^a \le Q^{\theta N}.$$

Thus

$$H(F) \le \log |\operatorname{range} F| \le N[h(\theta) + \theta \log Q].$$

The map F is deterministic, but this count gives the same bound for every distribution on $\mathcal{W}$. The proof of the finite theorem is complete.